

Harsh Pandey

Backend Development Enthusiast | AI/ML Learner | CSE (AI/ML) Student

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[LinkedIn](#) | [GitHub](#)

Profile

Second-year B.Tech (CSE – AI/ML) student with hands-on experience in Python-based backend development using FastAPI and SQLAlchemy. Comfortable working with relational databases and REST API design. Actively building full-stack project experience through academic and personal projects. Currently learning machine learning fundamentals and backend architecture patterns, with a focus on writing clean, structured, maintainable code.

Work Experience

Backend Developer — Academic & Personal Projects

2025 – Present

- Built REST APIs using FastAPI with modular routing and dependency injection for session and database management.
- Designed and implemented relational database schemas using SQLAlchemy ORM with SQLite.
- Structured projects into layered modules separating routes, models, schemas, and database logic.
- Practiced Git-based version control workflows including branching and collaborative code management via GitHub.

Education

B.Tech – Computer Science & Engineering (AI/ML)

2026 – Present

R. R. Institute of Modern Technology, Lucknow (AKTU) | Till 3rd Sem: 85.35%

High School (Class XII)

2022 – 2023

St. Xavier's School, Salempur, Deoria, UP | 65%

Secondary (Class X)

2020 – 2021

Cambridge International School, Mail Chauraha, Deoria, UP | 69%

Technical Skills

- **Languages:** Python, Java, C
- **Backend:** FastAPI, SQLAlchemy
- **Frontend:** HTML, CSS, JavaScript
- **Databases:** SQLite, PostgreSQL | MongoDB (Learning)
- **Python Libraries:** NumPy, Matplotlib
- **Core Concepts:** REST APIs, CRUD Operations, OOP, DBMS Basics, OS Fundamentals, Basic Data Structures
- **Tools:** Git, GitHub, VS Code
- **Currently Learning:** Machine Learning, Backend Architecture, AI/ML Fundamentals

Projects

Expense Tracker Backend API

2025

- Built a RESTful API using FastAPI with structured API routing and modular project layout.
- Defined relational database models using SQLAlchemy ORM and integrated persistent storage with SQLite.
- Implemented CRUD endpoints with dependency injection for database session management.
- Replaced in-memory data handling with a proper relational database layer.

PathForge – Roadmap Generator

2026

- Developed an AI-assisted platform for generating personalized learning roadmaps based on user goals.
- Designed a user-focused frontend interface with interactive roadmap planning and structured content flow.
- Built modular backend structure to support scalable feature additions and clean separation of concerns.
- Focused on personalization logic and dashboard usability as core design priorities.

HyperLapse Storyteller

2025

- Designed backend architecture for a structured content-flow platform focused on modular storytelling components.
- Planned API structure and data flow for sequential content delivery; currently under development.

Achievements & Exposure

- Participated in an AI-driven manufacturing optimization hackathon; contributed to problem analysis and predictive modeling ideation.
- Completed hands-on IoT project (ESP32-based autonomous obstacle-avoidance car) with sensor integration and state machine logic.

Certifications

- Introduction to Prompt Engineering for GenAI — LinkedIn Learning
- Secured Second Rank In Robotics
- Build With Your GenAI + Productivity Skills — Microsoft & LinkedIn
- Data Structures and Algorithms (Intermediate) — TrainX
- Blinking LED (IoT Hands-On) — Infosys Springboard
- Claude Code In Action – By Anthropic

Languages & Interests

Languages: English, Hindi | **Interests:** Backend Engineering, AI/ML Systems, Open Source Contribution, Stock Market Learning